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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEFSQ304 - SPICES AND OLEORESINS

(2024 ADMISSION ONWARDS)

Duration: 2 hours Maximum Marks: 70

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(1) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. (a) It helps delay spoilage and maintain food in a safer and acceptable condition. [U] / [CO1]
2. (b) Cuminaldehyde. [K] / [CO2]
3. (a) Eugenol. [K] / [CO3]
4. (a) Grinding → Steam distillation → Condensation → Separation → Filtration. [U] / [CO4]
5. (b) Turmeric. [K] / [CO3]
6. (a) They help delay oxidative deterioration and maintain food quality. [U] / [CO1]
7. (a) Moisture can affect both product stability and the weight used for pricing. [An] / [CO2]
8. (a) Threshing. [K] / [CO3]
9. (c) Extraction of desired flavour and aroma compounds. [K] / [CO4]
10. (a) Mature but green stage. [K] / [CO3]
11. (d) Cuminaldehyde. [K] / [CO3]
12. (b) Peeling. [U] / [CO3]
13. (b) Solvent extraction. [K] / [CO3]
14. (a) Polishing. [K] / [CO3]
15. (d) Steam or hydro-distillation. [K] / [CO3]
16. (a) Cinnamon. [A] / [CO3]
17. (b) Retained during drying. [U] / [CO3]
18. (a) Sulphur-containing compounds. [K] / [CO3]
19. (b) Winnowing. [K] / [CO3]
20. (c) Colour strength. [K] / [CO3]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

21. Describe the meaning and significance of spices in food preparation. [U] / [CO1]

Spices are plant-derived materials obtained from seeds, fruits, bark, roots, rhizomes, flower buds and other plant parts. They are used in small quantities to impart characteristic flavour, aroma, pungency and colour to food.

Significance of spices:

1. Flavour enhancement: Spices improve the taste and palatability of food. For example, black pepper provides pungency, while cinnamon gives a warm, sweet flavour.
2. Aroma development: Volatile compounds contribute to the characteristic aroma of foods. Eugenol in clove and cinnamaldehyde in cinnamon are examples.
3. Colour improvement: Turmeric provides yellow colour due to curcumin, while chilli provides red colour due to carotenoid pigments.
4. Food preservation: Certain spices possess antimicrobial and antioxidant properties that help delay microbial spoilage and oxidative deterioration.
5. Digestive benefits: Ginger, cumin and fennel are traditionally used to support digestion and relieve digestive discomfort.
6. Culinary identity: Different combinations of spices produce the characteristic flavours of regional and international cuisines.

22. List the major steps involved in the processing and manufacture of mace spice. [K] / [CO3]

Mace is the dried, lacy, reddish-orange aril surrounding the seed of the nutmeg fruit.

Major processing steps:

1. Harvesting: Mature nutmeg fruits are harvested when the outer fruit turns yellow and splits open naturally, before the fruits deteriorate.
2. Splitting: The outer fruit is removed or allowed to split open to expose the nutmeg seed surrounded by the reddish aril.
3. Aril removal: The mace aril is carefully separated from the nutmeg seed by hand.
4. Washing: The separated mace blades are washed in clean water to remove dirt and other impurities.
5. Blanching (optional): Mace may be blanched in hot water at approximately 80–90°C for 1–2 minutes to help retain its colour.
6. Drying: Mace is dried under the sun or in a mechanical dryer at approximately 40–50°C until the moisture content reaches about 8–10%.
7. Grading and sorting: The dried mace is sorted according to colour, size, length of the blades, appearance and cleanliness. Damaged or inferior pieces are removed.
8. Packaging: The dried mace is packed in airtight, moisture-proof packets or containers to protect it from moisture and contamination.
9. Storage: The packaged mace is stored in a cool, dry, well-ventilated place away from direct sunlight to preserve its aroma, colour and quality.

23. Identify the major by-products generated during oleoresin manufacture. [K] / [CO4]

Oleoresins are concentrated spice extracts containing both volatile oils and non-volatile resinous constituents responsible for the characteristic flavour, aroma, colour and pungency of spices.

Major by-products generated during oleoresin manufacture:

1. Spent spice (spent material): The solid residue remaining after solvent or supercritical CO_2 extraction. It may contain residual essential oils, fibre, carbohydrates and proteins.

2. Spent herbal biomass: The remaining plant material after extraction of herbs or spices. It can be used for composting, organic fertiliser production, bioenergy production and soil conditioning.

3. Recovered solvents: Organic solvents such as hexane, ethanol and acetone are recovered during evaporation or solvent-recovery operations. They can be purified and reused, reducing production costs and environmental pollution.
4. Carbon dioxide (CO_2): In supercritical CO_2 extraction, carbon dioxide is recovered after separation and recycled for repeated extraction cycles. This reduces operating costs and solvent-related waste.
5. Hydrosol (aromatic water): In processes involving steam or hydro-distillation, the aqueous layer obtained along with essential oil may contain dissolved aromatic compounds. Depending on its composition and quality, it may be used in suitable applications.
6. Dietary fibre: Spent spice residues may contain fibre and other plant components that can be evaluated for use in functional foods, bakery products and fibre supplements, subject to safety and suitability.
7. Natural pigments: Certain pigments may remain in the extraction residue, including curcuminoids from turmeric and capsanthin from chilli, depending on the extraction method and conditions.
8. Bioactive compounds: Residual polyphenols, flavonoids and other phytochemicals may remain in the spent material and could be recovered for suitable applications.
Utilisation of by-products:

- Spent spice residues and herbal biomass may be used for composting, organic fertiliser or bioenergy production.
- Recovered solvents and CO_2 can be recycled to reduce costs and waste.
- Suitable residual fibre, pigments and bioactive compounds may be recovered for further applications after quality and safety assessment.

Appropriate recovery and utilisation of oleoresin-processing by-products help minimise waste, improve resource efficiency and support sustainable spice processing.

24. Outline the major steps involved in the manufacture of spice oleoresins using supercritical CO_2 . [K] / [CO4]

Supercritical fluid extraction (SFE) is an advanced extraction method that uses carbon dioxide (CO_2) above its critical temperature and pressure to extract essential oils, oleoresins and other bioactive compounds from spices.
Major steps involved in the manufacture:
1. Selection of raw material: Select fresh, mature, aromatic spices such as black pepper, cardamom, ginger, turmeric or clove with good flavour and quality.
2. Cleaning: Remove foreign materials such as dust, stones, soil and plant debris to ensure the purity of the final product.
3. Drying: Dry the spices to an optimum moisture level to improve extraction efficiency and prevent microbial growth.
4. Grinding: Grind the dried spices to an appropriate particle size. This increases the surface area and improves the penetration of supercritical CO_2 .
5. Compression of CO_2 : Carbon dioxide is compressed using a high-pressure pump and heated above its critical temperature of approximately 31.1°C and critical pressure of 7.38 MPa (73.8 bar).
6. Formation of supercritical CO_2 : Under these conditions, CO_2 exhibits properties of both liquids and gases, enabling it to penetrate plant material and dissolve desirable compounds.
7. Extraction: Supercritical CO_2 is passed through the extraction vessel containing the prepared spice material. It dissolves essential oils, oleoresins, natural pigments, pungent compounds and other bioactive constituents.
8. Separation: The CO_2 carrying the dissolved compounds enters a separator. The pressure is reduced, decreasing the solubility of the extracted compounds and allowing the oleoresin to separate and be collected.
9. CO_2 recovery: The gaseous CO_2 is recovered, recompressed and recycled for subsequent extraction cycles, reducing operating costs and environmental impact.

10. Packaging and storage: The extracted oleoresin is collected and packed in suitable airtight containers, such as amber glass bottles or food-grade containers, and stored under cool, dry conditions away from direct light.

Advantages of supercritical CO_2 extraction:

- Produces high-purity extracts without organic solvent residues.
- Preserves heat-sensitive flavour and aroma compounds due to relatively low operating temperatures.
- Provides efficient and selective extraction of desirable constituents.
- Minimises oxidation and thermal degradation.
- CO_2 is recyclable, non-toxic and non-flammable under normal handling conditions.
- Produces high-quality essential oils and oleoresins suitable for food, pharmaceutical and cosmetic applications.

Limitations:

- Requires expensive high-pressure equipment.
- Needs specialised operation and maintenance.
- High initial investment may limit its use in small-scale industries.

25. Detail on the chemical composition of black pepper. [K] / [CO2]

Black pepper (Piper nigrum) is known as the "King of Spices." Its characteristic pungency and aroma are due to piperine and volatile oil constituents.

Chemical composition of black pepper:

1. Piperine: The major pungent alkaloid responsible for the characteristic sharp taste of black pepper. It contributes to the spice's commercial and functional value.
2. Volatile oils: These provide the characteristic aroma and flavour. Important constituents include pinene, limonene and caryophyllene.
3. Fixed oils: Black pepper contains non-volatile lipids that contribute to its chemical composition.
4. Resinous substances: These contribute to the flavour and other characteristics of pepper extracts and oleoresins.
5. Carbohydrates: Include starch and other carbohydrates that contribute to the bulk of the spice.
6. Proteins: Present in smaller quantities and contribute to its overall nutritional composition.
7. Dietary fibre: Derived from the structural components of the dried fruit.
8. Minerals and moisture: Black pepper contains mineral constituents, while its moisture content influences storage stability and quality.

Piperine is mainly responsible for pungency, while volatile oils contribute to the characteristic aroma. The composition varies with variety, maturity, processing and storage conditions.

26. Recall the importance of curing in vanilla processing. [K] / [CO3]

Curing is a controlled post-harvest process through which harvested vanilla beans develop their characteristic brown colour, aroma, flavour and texture. It involves four major stages: killing, sweating, drying and conditioning.

Importance of curing:

1. Vanillin formation: Enzymatic hydrolysis converts glucovanillin into vanillin, the principal compound responsible for the characteristic flavour and aroma of vanilla.
2. Flavour development: Curing develops the characteristic sweet, pleasant and complex flavour of vanilla. Sweating is particularly important for flavour development.
3. Colour development: The green beans gradually develop their characteristic brown colour during curing.
4. Moisture reduction: Controlled drying reduces moisture content, helping prevent fungal growth and spoilage.
5. Texture improvement: Proper curing produces flexible beans without making them excessively brittle.

6. Aroma maturation: Conditioning or ageing allows flavour compounds to mature, producing a richer, smoother and more complex aroma.
7. Improved market value: Properly cured beans have superior sensory quality, better appearance and greater commercial value.

Four major stages of vanilla curing:

- Killing: Stops normal physiological activities while preserving enzymes responsible for flavour development.
- Sweating: Promotes enzymatic activity, vanillin formation, brown colour development and aroma production.
- Drying: Gradually reduces moisture while maintaining flexibility and preventing spoilage.
- Conditioning (ageing): Allows flavour compounds to mature and stabilises the quality of cured beans.

27. Explain the importance of harvesting cardamom capsules at the appropriate stage. [U] / [CO3]

Cardamom capsules should be harvested when they are fully mature but still green, before they split open naturally.

Importance of harvesting at the correct stage:

1. Maximum essential oil content: Fully mature capsules contain well-developed seeds with a high concentration of volatile oils responsible for the characteristic aroma and flavour.
2. Better flavour and aroma: Harvesting at the appropriate maturity ensures the development of the desirable cardamom flavour and aroma.
3. Prevention of seed loss: Harvesting before the capsules split prevents seeds from falling out, thereby reducing losses.
4. Colour retention: Mature green capsules retain their attractive green colour after proper curing, improving their appearance and quality.
5. Improved market value: Uniform, green and aromatic capsules receive better commercial acceptance and higher market value.
6. Reduced processing losses: Correct harvesting improves recovery of high-quality capsules and reduces losses caused by immature or overripe produce.

Harvesting method:

- Cardamom capsules do not mature uniformly on the plant.
- Therefore, harvesting is carried out manually at regular intervals, usually every 15–30 days, depending on the season and variety.
- Only mature capsules are collected, while immature capsules are left on the plant for further development.

Harvesting at the correct stage ensures maximum yield, good essential oil content, attractive green colour and superior commercial quality.

28. Explain the need for careful handling of saffron after harvesting. [U] / [CO3]

Saffron is obtained from the dried stigmas of Crocus sativus. It is a delicate and expensive spice; therefore, careful post-harvest handling is essential to preserve its quality.

Major post-harvest operations and their importance:

1. Flower collection: Flowers are collected carefully, preferably during the cool morning hours, to minimise damage and preserve freshness.
2. Stigma separation: The red stigmas are separated manually from the flowers without damaging or mixing them with unwanted floral parts.
3. Drying: The stigmas are dried carefully under controlled conditions to reduce moisture and prevent microbial spoilage.
4. Colour preservation: Careful drying and protection from excessive heat and light help preserve crocin, the principal colouring compound.
5. Aroma preservation: Appropriate handling helps retain safranal and other volatile compounds responsible for saffron’s characteristic aroma.

6. Cleaning and grading: Foreign matter is removed, and saffron is graded according to colour, aroma, purity and quality.
7. Packaging and storage: Saffron is packed in airtight, moisture-resistant containers and stored in a cool, dark and dry place.

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

29. Define oleoresins and describe their chemical composition. Explain the role of volatile and non-volatile constituents in determining the characteristic properties of oleoresins. [U] / [CO4]

Definition

Oleoresins are concentrated, semi-solid or viscous extracts obtained from spices and aromatic plant materials, generally by solvent extraction. They contain both volatile essential oils and non-volatile resinous constituents, providing the characteristic aroma, flavour, colour and pungency of the original spice.

Chemical composition of oleoresins

Oleoresins are complex mixtures of volatile and non-volatile organic compounds.

1. Volatile fraction (essential oil fraction)

This fraction contains volatile aromatic compounds responsible mainly for the characteristic aroma and flavour of spices.

- Monoterpenes: Examples include limonene, α-pinene, β-pinene and sabinene. They contribute fresh, citrus-like and herbal aromas.
- Sesquiterpenes: Examples include caryophyllene, zingiberene and humulene. They contribute warm, spicy, woody and earthy aromas.
- Oxygenated terpenoids: These include alcohols, aldehydes, ketones, phenols, esters and oxides that contribute to aroma, flavour and biological activity.

2. Non-volatile fraction (resinous fraction)

This fraction contains non-volatile compounds responsible for the body, colour, pungency and persistence of flavour.

- Resins: High-molecular-weight compounds that provide viscosity, body and flavour retention.
- Natural pigments: Curcumin in turmeric, capsanthin and capsorubin in paprika, and chlorophyll in herbs contribute characteristic colours.
- Pungent principles: Piperine in black pepper, capsaicin in chilli, and gingerol in fresh ginger contribute characteristic pungency.
- Fixed oils: Non-volatile triglycerides present in certain spices, such as nutmeg, coriander and black pepper, contribute to the composition and stability of oleoresins.
- Waxes: Present in small quantities, they protect plant tissues and increase the viscosity and stability of oleoresins.

Role of volatile and non-volatile constituents

A. Volatile constituents

1. Aroma: Provide the characteristic smell of the spice.
2. Flavour: Contribute to the overall sensory profile of the extract.
3. Freshness: Their retention helps maintain the desirable aroma and flavour.
4. Biological activity: Certain compounds possess antimicrobial and antioxidant properties.

B. Non-volatile constituents

1. Pungency: Compounds such as piperine and capsaicin provide the characteristic pungent sensation.
2. Colour: Natural pigments contribute the characteristic colour of the oleoresin.
3. Flavour persistence: Resinous constituents help retain flavour for a longer period.
4. Body and viscosity: Resins contribute to the concentrated, viscous nature of oleoresins.

5. Stability: Resinous compounds improve flavour retention and contribute to greater chemical stability during storage.

The volatile fraction primarily determines the aroma and contributes to flavour, whereas the non-volatile fraction contributes to pungency, colour, viscosity, flavour persistence and stability. Together, these fractions provide the complete sensory and functional properties of oleoresins.

30. Explain the manufacture of two different value-added coriander products. [U] / [CO3]

Coriander (Coriandrum sativum) is an important seed spice valued for its characteristic flavour and aroma. Value addition improves its convenience, shelf life, commercial value and industrial applications.

1. Coriander powder

Coriander powder is produced by cleaning, drying and grinding coriander seeds.

Steps involved:

1. Harvesting: Coriander plants are harvested when the fruits reach physiological maturity.
2. Threshing: The dried or harvested plant material is threshed to separate the coriander fruits from the stems.
3. Cleaning: Dust, stones, stalks and other foreign materials are removed.
4. Drying: The seeds are dried to a suitable moisture level to improve storage stability and prevent mould growth.
5. Grading: Seeds are graded based on size, colour, cleanliness and quality.
6. Roasting (optional): The seeds may be lightly roasted to enhance their characteristic aroma.
7. Grinding: The seeds are ground into a fine or coarse powder, depending on the intended application.
8. Sieving: The powder is sieved to obtain a uniform particle size.
9. Packaging: The powder is packed in airtight, moisture-resistant packaging to minimise loss of volatile oils.

Uses:

- Used in curry powders, spice blends, sauces and pickles.
- Adds a mild, sweet and citrus-like flavour to foods.
- Used in meat products, soups and vegetable preparations.

2. Coriander oleoresin

Coriander oleoresin is a concentrated extract containing volatile and non-volatile constituents responsible for the characteristic flavour and aroma of coriander.

Steps involved:

1. Cleaning: Coriander seeds are cleaned to remove foreign matter.
2. Drying: The seeds are dried to a suitable moisture level.
3. Grinding: The seeds are ground to an appropriate particle size to facilitate extraction.
4. Solvent extraction: A suitable food-grade solvent is passed through or mixed with the ground seeds to extract the desirable compounds.
5. Filtration: The extract is filtered to remove suspended solid particles.
6. Solvent recovery: The solvent is removed, usually under controlled conditions, and recovered for reuse where appropriate.
7. Oleoresin collection: The concentrated extract is collected.
8. Packaging and storage: The product is packed in suitable airtight containers and stored under appropriate conditions.

Uses:

- Used in processed foods, sauces, seasonings and spice blends.
- Provides consistent flavour in industrial food formulations.
- Used in food, pharmaceutical and cosmetic applications.

Coriander powder provides a convenient spice in powdered form, whereas coriander oleoresin provides a concentrated flavouring ingredient suitable for standardised industrial applications.

31. Describe how spices contribute to the sensory quality and acceptability of foods through their effects on flavour, taste, aroma, texture and colour. [U] / [CO1]

Spices contribute significantly to the sensory quality and consumer acceptability of foods. Their essential oils, pigments, pungent compounds and other bioactive constituents influence the appearance, smell, taste and mouthfeel of food products.

1. Flavour enhancement
    - Spices impart distinctive flavour characteristics and improve the palatability of food.
    - Different spices provide sweet, warm, earthy, pungent or bitter flavour notes.
    - Black pepper provides a sharp flavour due to piperine, while cinnamon provides a warm, sweet flavour due to cinnamaldehyde.
    - Spice combinations produce distinctive flavours in regional and international cuisines.
2. Taste and pungency
    - Spices stimulate different taste and oral sensory responses.
    - Piperine in black pepper and capsaicinoids in chilli produce pungency.
    - Ginger contributes a warm, spicy sensation due to gingerols and related compounds.
    - Appropriate amounts of spices enhance the overall eating experience, whereas excessive amounts may cause undesirable intensity.
3. Aroma development
    - Volatile compounds in spices contribute to the characteristic aroma of foods.
    - Eugenol in clove provides a strong, warm aroma.
    - Cinnamaldehyde gives cinnamon its characteristic fragrance.
    - Linalool contributes to the pleasant citrus-like aroma of coriander.
    - Aroma strongly influences appetite and consumer perception of food quality.
4. Colour improvement
    - Natural pigments in spices improve the visual appearance of food.
    - Curcumin in turmeric imparts a yellow colour.
    - Capsanthin and capsorubin in chilli contribute red colour.
    - Saffron imparts a golden-yellow colour due to crocin.
    - Attractive colour increases visual appeal and can improve consumer acceptance.
5. Texture and mouthfeel
    - Certain spices and spice ingredients influence the physical characteristics of food.
    - Mustard seeds contribute texture to pickles, sauces and condiments.
    - Sesame seeds add crunchiness to bakery products and confectionery.
    - Spice powders can contribute body or thickness to some formulations, depending on the product and quantity used.
6. Improvement of overall acceptability
    - Spices combine flavour, aroma, colour and mouthfeel to create a balanced sensory profile.
    - They can make bland foods more appealing.
    - Suitable spice combinations improve the identity and uniqueness of food products.
    - Correct proportions are essential to avoid excessive pungency, bitterness or overpowering aroma.
7. Contribution to shelf life
    - Some spices contain antioxidant and antimicrobial compounds that help delay oxidative deterioration and microbial spoilage.
    - Maintaining food quality during storage can help preserve its sensory acceptability.

Spices improve the sensory quality of food through their combined effects on flavour, taste, aroma, texture and colour. Proper selection and use of spices increase palatability and consumer acceptance.

32. Explain the dietary and medicinal uses of cinnamon, clove, fennel, fenugreek, nutmeg, mace and saffron. [U] / [CO2]

Spices are used in foods and beverages to improve flavour, aroma, colour and palatability. Many also contain bioactive compounds associated with antioxidant, antimicrobial, digestive and other biological activities. Their medicinal uses should be distinguished from proven clinical treatments.

1. Cinnamon

Active compound: Cinnamaldehyde.

Dietary uses:

- Used in cakes, biscuits, chocolates, desserts and confectionery.
- Added to beverages, breakfast cereals and spice blends.
- Provides a warm, sweet aroma and characteristic flavour.

Medicinal significance:

- Contains antioxidant compounds.
- Exhibits antimicrobial activity in laboratory studies.
- Has been studied for potential effects on blood glucose regulation.
- Traditionally used to support digestion.

2. Clove

Active compound: Eugenol.

Dietary uses:

- Used in spice blends, bakery products, pickles and beverages.
- Added to meat preparations, sauces and confectionery.
- Provides a strong, warm and aromatic flavour.

Medicinal significance:

- Exhibits antioxidant and antimicrobial activity.
- Traditionally used for digestive discomfort.
- Clove oil is used in dentistry for its local analgesic properties under appropriate professional guidance.

3. Fennel

Active compound: Anethole.

Dietary uses:

- Used as a mouth freshener after meals.
- Added to confectionery, bakery products and beverages.
- Used in spice mixtures and traditional digestive preparations.

Medicinal significance:

- Traditionally used as a digestive and carminative agent.
- May help relieve mild flatulence and bloating.
- Its aromatic compounds contribute to its characteristic flavour and digestive use.

4. Fenugreek

Active constituents: Sotolon, saponins, alkaloids and mucilage.

Dietary uses:

- Used in curry powders, pickles and traditional dishes.
- Added to spice blends and selected functional food products.
- Provides a characteristic bitter taste and strong aroma.

Medicinal significance:

- Contains dietary fibre and other bioactive constituents.
- Has been studied for potential effects on blood glucose regulation.
- Has also been investigated for its possible role in cholesterol management.
- Traditionally used to support digestion.

5. Nutmeg

Active constituents: Myristicin, sabinene and pinene.

Dietary uses:

- Used in cakes, pastries, puddings and confectionery.
- Added to dairy products, sauces, soups and meat preparations.

- Provides a warm, sweet and slightly nutty flavour.

Medicinal significance:

- Traditionally used as a digestive aid.
- Contains compounds with antioxidant activity.
- Exhibits antimicrobial activity in laboratory studies.

Caution: Nutmeg should be used in normal culinary quantities because excessive consumption can cause toxicity.

6. Mace

Active constituents: Myristicin and other volatile aromatic compounds.

Dietary uses:

- Used in spice mixtures, bakery products and confectionery.
- Added to sauces, soups and meat preparations.
- Provides a warm, delicate and aromatic flavour.

Medicinal significance:

- Traditionally used to support digestion.
- Contains aromatic compounds with antioxidant activity.
- Its volatile constituents have been investigated for potential biological activities.

7. Saffron

Active constituents: Crocin, picrocrocin and safranal.

Dietary uses:

- Used in rice dishes, desserts, sweets and confectionery.
- Added to milk-based preparations and beverages.
- Crocin provides golden-yellow colour, picrocrocin contributes bitterness, and safranal contributes characteristic aroma.

Medicinal significance:

- Contains antioxidant compounds, particularly crocin.
- Has been studied for potential effects on mood and other aspects of health.
- Traditionally used in certain preparations to support general well-being.

Cinnamon, clove, fennel, fenugreek, nutmeg, mace and saffron have important dietary applications because of their characteristic flavour, aroma and colour. Their bioactive constituents also contribute to their traditional medicinal uses and are subjects of scientific research. However, these spices should not replace prescribed medicines or established medical treatments.